

TITLE: STM32 BASED TEMPERATURE MONITORING AND ALERT SYSTEM

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1.0 INTRODUCTION

This project is about a temperature monitoring system using STM32 Nucleo board. The system is designed to monitor temperature level and give alert when temperature goes high. It uses three LEDs to show status and a buzzer for critical condition.

2.0 OBJECTIVE

The main objective is to design a system that can:

- Read temperature from potentiometer as sensor
- Show SAFE, WARNING and CRITICAL states
- Turn on buzzer when critical
- Have a mute button to off the buzzer

3.0 COMPONENTS USED

- STM32 Nucleo C031C6 Board x1
- Potentiometer (as temperature sensor) x1
- LEDs (Green, Yellow, Red) x3
- Buzzer x1
- Push button x1
- Jumper wires

Software used

- Wokwi Online Simulator
- Arduino IDE

4.0 CIRCUIT CONNECTION

The potentiometer VCC was connected to 5V, GND to GND and signal to A0 of the STM32. The Green LED was connected to D2, Yellow to D3, Red to D4. Buzzer was connected to D5 and push button to D6. All grounds were common.

Wokwi was used for simulation because the physical STM32 board was not available.

5.0 HOW THE SYSTEM WORKS

The system reads analog value from A0 (0-1023). This value represents temperature.

- When value < 30: SAFE state, only Green LED ON
- When value 30 to 70: WARNING state, only Yellow LED ON
- When value > 70: CRITICAL state, Red LED ON and buzzer ON

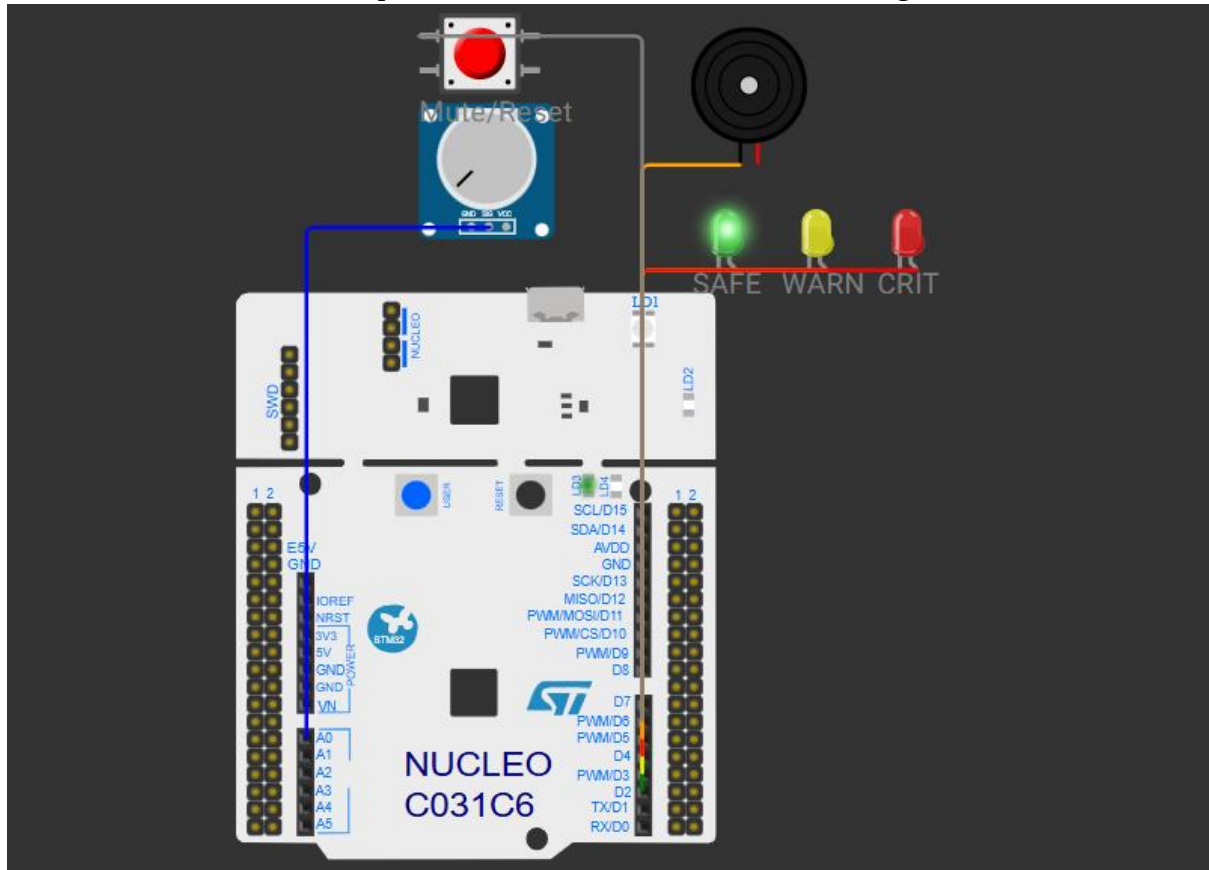
When push button is pressed during CRITICAL, the buzzer will mute for 5 seconds then

come back if temperature still high.

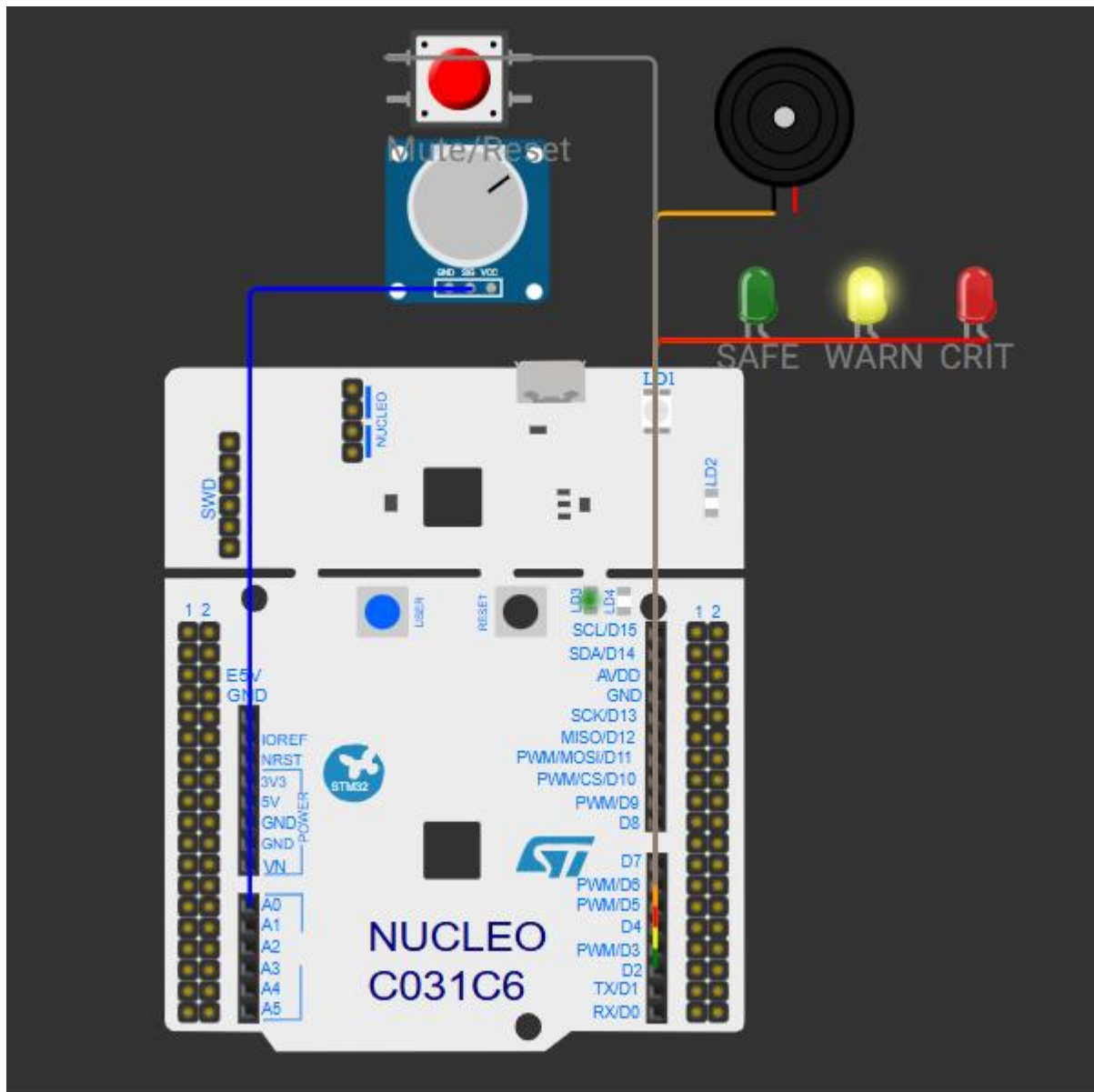
6.0 RESULTS

The simulation was tested in Wokwi:

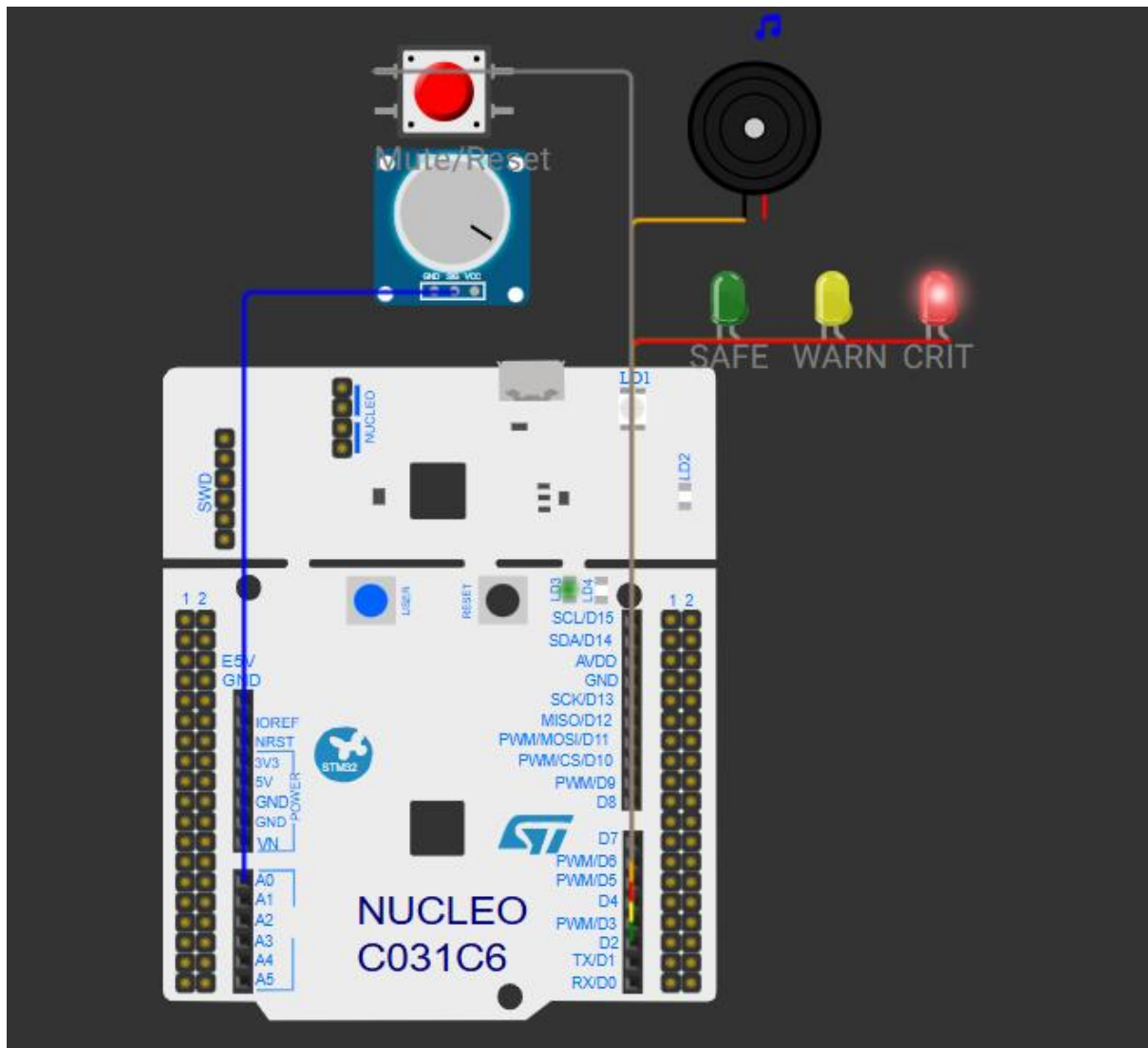
Test 1: Potentiometer at low position - Green LED came ON, meaning SAFE.



Test 2: Potentiometer at middle - Yellow LED came ON, meaning WARNING.



Test 3: Potentiometer at high - Red LED and buzzer came ON, meaning CRITICAL.



The state transitions worked as expected.

7.0 CHALLENGES

During simulation, it was observed that the Nucleo C031C6 board in Wokwi does not show Serial Monitor output. The status could only be verified through the LEDs. On real STM32 hardware, `Serial.println()` would print through USART2 via ST-Link in STM32CubeIDE.

Also finding the board in Wokwi was difficult because some STM32 boards have been removed from the platform.

8.0 CONCLUSION

The temperature monitoring system was successfully designed and simulated. All three states were working. The system can be used in real life applications like room temperature monitoring, server room or industrial machines where overheating is dangerous.

9.0 CODE AND WOKWI LINK

<https://github.com/lanafriyie/zapspace-internship.git>

10.0 REFERENCES

STMicroelectronics, "STM32 Nucleo-32 board User Manual (UM1956)," STMicroelectronics, 2023.[1]

Wokwi, "Wokwi Online Simulator Documentation," Available: <https://docs.wokwi.com>, Accessed: Sept. 2026.[2]